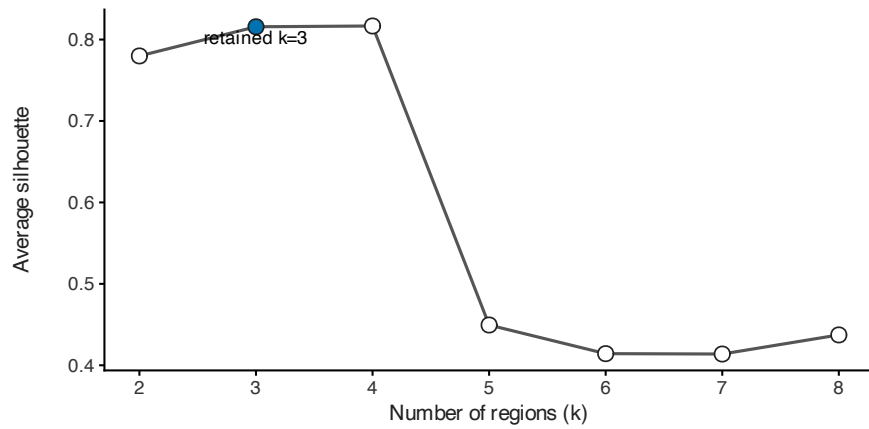


A

Primary warm-start-region selection

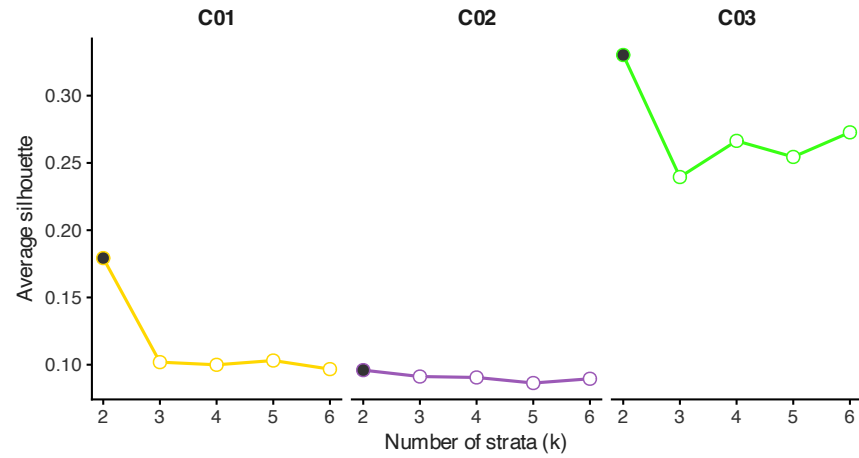
k=3 and k=4 differ by <0.001; k=4 has a singleton.
80% subsample re-selection: 58/100 versus 42/100.



B

Within-region warm-start strata

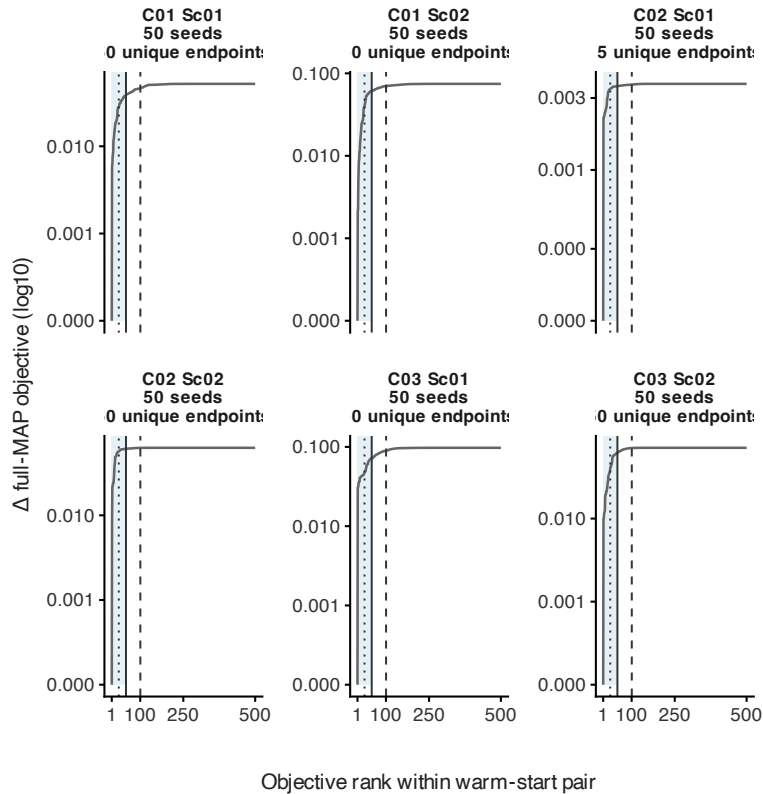
k=2 maximizes silhouette in each region.
Low values limit interpretation to search coverage.



C

Pair-specific full-MAP objective eligibility

Blue region: primary lowest 10% (50/500); lines mark 5%, 10%, and 20%.
Exact zero is displayed at $1e-4$.



D

Seed-, cutoff-, and endpoint-multiplicity robustness

Primary 10% modal result/support; 'all q' = unchanged at 5%, 10%, and 20%.
Red cross = changed after exact-parameter endpoint deduplication.

